

2.13 OTHER ENGINE AND TRANSMISSION LIMITATIONS

2.13.1 Engine Oil Pressure

Below 79% N_1 speed 3.5 to 9.1 kp/cm²

From 79 to 94% N_1 speed 6.3 to 9.1 kp/cm²

At 94% N_1 speed and above 8.1 to 9.1 kp/cm²

During engine starting, a positive oil pressure indication must be obtained upon reaching 59% N_1 (Idle) speed.

NOTE During cold weather operation, 10 kp/cm² engine oil pressure is allowable following an engine start. When the 9.1 kp/cm² limit is exceeded, operate engine at minimum power (IDLE) until normal oil pressure limits are attained.

2.13.2 Engine Oil Temperature

Minimum during start -30 °C

Normal operating range +30 to +107 °C

NOTE Engines may be operated at oil temperatures from 0 to 107 °C provided engine oil pressure is within specified limits.

EFFECTIVITY *If extreme low temperature kit 105-80006 installed and operational*

Normal operating range 0 to +107 °C

EFFECTIVITY ALL

Maximum +107 °C

2.13.3 Engine Starter

To prevent starter overheat damage, limit starter energize time to the following:

External Power	Onboard Battery
20 seconds - ON	30 seconds - ON
60 seconds - OFF	60 seconds - OFF
20 seconds - ON	30 seconds - ON
60 seconds - OFF	60 seconds - OFF
20 seconds - ON	30 seconds - ON
30 minutes - OFF	30 minutes - OFF

The 30-minute cooling period is required before beginning another starting cycle.

The starter energize time is the time that elapses between initiation of the starter and ignition in the turbine.

Motoring condition

External Power	Onboard Battery
15 seconds - ON	20 seconds - ON
60 seconds - OFF	60 seconds - OFF
15 seconds - ON	20 seconds - ON
23 minutes - OFF	23 minutes - OFF

2.13.4 Generator

Maximum load per generator 150 A

2.13.5 Ground Power Starts

The current flow during starting engines shall not exceed 600 A when using 28V DC ground power units.

2.13.6 Main Transmission Oil Pressure

Minimum 0.5 kp/cm²

Normal range 0.8 to 2.0 kp/cm²

NOTE During engine starting, the oil pressure may exceed the normal range.